

Smart Study Difficulty Predictor (AI Project)

Fundamentals of AI and Machine Learning (CSA2001)

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1. INTRODUCTION

Artificial Intelligence (AI) and Machine Learning (ML) are widely used to solve real-world problems by analysing data and making predictions. In the education domain, students often struggle to identify which subjects or topics require more attention.

This project presents an AI-based solution to predict the difficulty level of study conditions using machine learning techniques.

2. Problem Statement

Students often:

- Do not know which subjects are difficult
- Spend time inefficiently
- Lack personalized study guidance

This leads to poor academic performance and ineffective study planning.

3. Proposed Solution

The solution is a **Smart Study Difficulty Predictor** that:

- Takes student-related inputs (study hours, attendance, etc.)
- Uses a machine learning model
- Predicts whether the study condition is:
 - Easy
 - Medium
 - Hard

4. Methodology

Step 1: Data Collection

A dataset was created manually with features:

- Hours studied
- Attendance
- Previous score
- Sleep hours

Step 2: Data Preprocessing

- Data loaded using Pandas
- Feature scaling applied using StandardScaler

Step 3: Model Selection

- Algorithm used: **Logistic Regression**
- Type: Supervised Learning (Classification)

Step 4: Training

- Dataset split into training and testing sets
- Model trained using scikit-learn

Step 5: Prediction

- User input is taken
- Model predicts difficulty level

5. Algorithm Used

Logistic Regression

- Used for classification problems

- Predicts probability of classes
 - Suitable for categorical outputs
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6. Implementation

Tools Used:

- Python
- Pandas
- Scikit-learn
- Pickle

Files:

- `train_model.py` → trains model
 - `predict.py` → predicts output
 - `student_data.csv` → dataset
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7. Sample Output

```
Enter study hours: 4
Enter attendance: 75
Enter previous score: 60
Enter sleep hours: 6
Predicted Difficulty: Medium
```

8. Limitations

- Dataset is small and manually created
 - Model may not generalize well to real-world data
 - Predictions may be inaccurate for extreme values
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9. Future Scope

- Use real student datasets
 - Add a web interface (Streamlit)
 - Use advanced models (Random Forest, Neural Networks)
 - Provide personalized study plans
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10. Course Relevance

This project aligns with the course syllabus:

- Supervised Learning (CO4)
 - Machine Learning Algorithms (CO2, CO3)
 - Real-world Problem Solving (CO5)
 - Data Processing and Feature Handling
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11. Conclusion

This project demonstrates how machine learning can be used to solve a real-world problem in education. It helps students understand their study patterns and improve their performance through data-driven insights.
